

# ALI RAZA CHAUDHRY

AI/ML Engineer | Data Scientist | Computer Vision | Generative AI | Full Stack Developer

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## PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with hands-on experience in machine learning, deep learning, computer vision, data analytics, statistical modeling, and generative AI. Skilled in developing end-to-end AI solutions, from data preparation and model development to evaluation, optimization, and deployment. Proficient in Python, PyTorch, TensorFlow, OpenCV, YOLO, R, SQL, LLMs, RAG, and AI agents, with full-stack experience in FastAPI, React, Next.js, and Node.js. Focused on building scalable, data-driven, and production-ready AI solutions that deliver measurable results.

## EXPERIENCE

### Embedded Systems Intern | Meta Desk Global — Lahore, Pakistan

July 2025 – July 2026

- Worked on AI/ML, Data Science, Computer Vision, and NLP tasks involving data preprocessing, model development, evaluation, and optimization.
- Contributed to microcontroller programming, circuit interfacing, hardware debugging, and embedded system deployment on Orange Pi platforms.

### Freelance Computer Vision Engineer | Self-Employed / Fiverr

2024 – Present

- Built YOLO-based computer vision solutions with FastAPI deployment.
- Built a golf swing tracking system using pose estimation to analyze swing angles and tempo for coaching feedback.

## EDUCATION

Bachelor of Computer Engineering | COMSATS University, Lahore | CGPA: 2.81/4.00

2022 – 2026

F.Sc. Pre-Engineering | Govt. Islamia College Civil Lines, Lahore

2020 – 2022

## PROJECTS & PORTFOLIO

### UFC Vision AI — Real-Time Fight Analytics

- Custom YOLOv11 fighter-tracking and action-detection (punches, kicks, grappling) with pose estimation, fight stats, and a React dashboard.

### AI Cricket Match Analytics System

- YOLOv8 + OpenCV pipeline for player/ball detection, field mapping, shot analysis, pitch maps, and automated highlight generation.

### Smart Football Match Analysis System

- Broadcast-footage analytics: player/ball tracking, team classification, bird's-eye-view transform, speed estimation, heatmaps, event detection.

### AI Skeleton Detection & 3D Avatar Generation

- Real-time MediaPipe Pose pipeline (30+ FPS), 3D avatar generation with Trimesh/Open3D, coach-correction blending, FastAPI integration.

### Smart Vehicle Gate Access System (Final Year Project)

- License-plate detection and vehicle make/model recognition (VMMR) for automated, all-weather gate access. | [automated-gate-access.vercel.app](https://automated-gate-access.vercel.app)

### AI Baseball Swing Performance Analyzer

- CV pipeline for batting video: pose estimation, swing-phase and strike-zone detection, bat/ball tracking, biomechanics scoring.
- MediaPipe Pose, YOLO, and GPU-accelerated inference with annotated videos and a Streamlit dashboard.

### Additional Computer Vision + Data Science Projects

- **Automatic Product Scanning & Billing System** — vision-based retail checkout automation.
- **Real-Time Multi-Person Tracking System** — multi-object tracking with analytics output.
- **Automatic Vehicle Counting System** — real-time traffic counting and analytics.

- **Vehicle Damage Detection and Cost Estimator** — damage localization with automated cost estimation.
- **Real-Time Tools & People Detection** — object detection for safety and analytics.
- **Waste Classification & Furniture Classification** — visual sorting and category recognition.
- **Bayesian Statistical Analysis & Predictive Modeling** — posterior estimation and hypothesis testing on real-world data.

## Full-Stack Web Applications

- **Crown and Dial** — Luxury e-commerce storefront (React). | [crown-and-dial.netlify.app](https://crown-and-dial.netlify.app)
- **Fine Crafted Structures** — Luxury construction company site (Next.js). | [constractioncompany.netlify.app](https://constractioncompany.netlify.app)
- **Healthy Start** — Non-Emergency Medical Transportation platform (WordPress/PHP). | [healthystartnc.com](https://healthystartnc.com)
- **Nyuton Enterprises** — Build Real Cashflow (WordPress/PHP). | [nyutonenterprises.com](https://nyutonenterprises.com)
- **Stepping Stones CRI** — Mental Health, Well-being & Prevention (React). | [steppingstone.netlify.app](https://steppingstone.netlify.app)
- **Revara** — Real estate investment platform (Next.js). | [hotel-web-chi-tan.vercel.app](https://hotel-web-chi-tan.vercel.app)
- **NFT Marketplace** — Discover, create & trade extraordinary NFTs (React.js). | [nftmarketplaece.netlify.app](https://nftmarketplaece.netlify.app)
- **Techlo** — Cutting-edge online solutions agency (React.js). | [techloagency.netlify.app](https://techloagency.netlify.app)
- **Clipzy** — AI-powered live streaming & content studio (Next.js, React, Node.js, AI APIs). | [clipzy.xynovix.com](https://clipzy.xynovix.com)
- **FamLink** — Family care platform connecting families with caregivers. | [famlink.care](https://famlink.care)
- **NOVA — Premium Footwear** — Interactive 3D web experience with real-time rendering. | [core-seven-henna.vercel.app](https://core-seven-henna.vercel.app)
- **Luxury Watch Store** — Modern e-commerce storefront (Next.js, React). | [watches-gray.vercel.app](https://watches-gray.vercel.app)

## TECHNICAL SKILLS

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**Computer Vision:** OpenCV, MediaPipe, YOLOv8/v11, RF-DETR, Object Detection, Image Classification, Image Segmentation, Multi-Object Tracking, OCR, Real-Time Vision Pipelines

**Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Decision Trees, Random Forest, XGBoost, LightGBM, SVM, KNN, Naive Bayes, Ensemble Learning, Dimensionality Reduction, PCA, Feature Selection, Model Evaluation & Optimization

**Deep Learning:** PyTorch, TensorFlow, Keras, CNNs, RNNs, LSTMs, GRUs, Autoencoders, GANs, Transfer Learning, Attention Mechanisms, Transformers, Neural Networks

**Generative AI & LLMs:** Generative AI, Large Language Models (LLMs), Prompt Engineering, RAG, Vector Databases, Embeddings, Transformers, AI Agents, Agentic AI, LangChain, LangGraph, Hugging Face, Fine-Tuning, LLM APIs

**Data Science & Analytics:** Data Analysis, Data Analytics, EDA, Data Cleaning, Data Preprocessing, Feature Engineering, Predictive Analytics, Data Visualization, Experiment Tracking

**Statistical Analysis:** Descriptive & Inferential Statistics, Hypothesis Testing, t-tests, ANOVA, Chi-Square Tests, Correlation & Regression, Probability, Confidence Intervals, Bayesian Analysis, A/B Testing

**Statistical & Analytics Tools:** R, RStudio, Stata, Python, Jupyter, Power BI, Tableau, Excel

**Python Data Libraries:** NumPy, Pandas, SciPy, Statsmodels, Scikit-learn, Matplotlib, Seaborn, Plotly

**Languages & Backend:** Python 3.9+, FastAPI, JavaScript (ES6+), Node.js, Express.js

**Frontend:** React, Next.js, HTML, CSS, Tailwind, Bootstrap

**Databases:** SQL, PostgreSQL, SQL Server, MySQL, MongoDB, NoSQL

**Tools:** Git, GitHub, VS Code, Postman, REST APIs, Webhooks

**Cloud & SaaS:** AWS, SaaS Platforms, Docker, Kubernetes, CI/CD, Linux

## CERTIFICATIONS

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- Python Machine Learning — Udemy
- OpenCV Bootcamp — OpenCV University
- Artificial Intelligence — UniAthena